CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

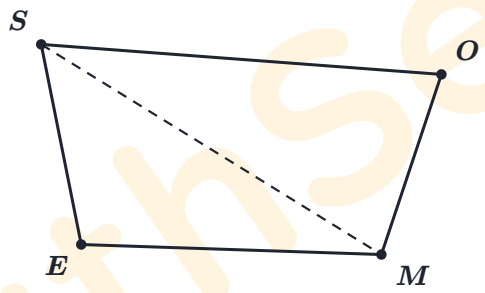
Chapter 4: Quadrilaterals

Topic-wise short notes with solved examples

Rectangles, squares, parallelograms, rhombuses, kites and trapeziums: their definitions, the properties of their sides, angles and diagonals, and how these properties follow from congruent triangles and parallel lines.

1 Quadrilaterals and their angles

- A **quadrilateral** is a closed figure with four sides. It has four angles and two diagonals.
- **The sum of the angles of any quadrilateral is** $360°$. A diagonal cuts it into two triangles, and each has an angle sum of $180°$.



Diagonal SM divides the quadrilateral into two triangles

- So a quadrilateral cannot have exactly three right angles: the fourth is forced to be $360° - 270° = 90°$.

Example 1.1 Three angles of a quadrilateral are $85°$, $110°$ and $75°$. Find the fourth.

SOLUTION

$360° - (85° + 110° + 75°) = 360° - 270°$

$90°$

Example 1.2 The angles of a quadrilateral are x, $2x$, $3x$ and $4x$. Find them.

SOLUTION

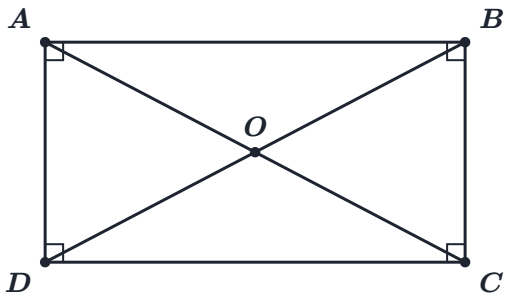
$10x = 360°$, so $x = 36°$

$36°, 72°, 108°, 144°$

2 Rectangle

- A **rectangle** is a quadrilateral in which all the angles are $90°$.

- **Properties**: opposite sides are equal; opposite sides are parallel; the diagonals are **equal** and **bisect each other**.
- Why the diagonals are equal: in rectangle ABCD, triangles ADC and DAB have $DC = AB$, $\angle D = \angle A = 90^\circ$ and AD common. They are congruent (SAS), so $AC = BD$.
- **Converse**: if the diagonals of a quadrilateral are equal and bisect each other, it is a rectangle, whatever the angle between them.



Rectangle: OA = OB = OC = OD

- Since $OA = OB = OC = OD$, the four triangles formed by the diagonals are isosceles.

Example 2.1 In rectangle ABCD the diagonals meet at O and $\angle AOB = 110^\circ$. Find $\angle OAB$ and $\angle OAD$.

SOLUTION

Triangle AOB is isosceles with $OA = OB$, so $\angle OAB = \dfrac{180^\circ - 110^\circ}{2} = 35^\circ$.

$\angle OAD = 90^\circ - 35^\circ$

$\angle OAB = 35^\circ$ **and** $\angle OAD = 55^\circ$

Example 2.2 A carpenter joins two strips of wood at their mid-points. One strip is 10 cm long. What must be true for the four ends to be the corners of a rectangle?

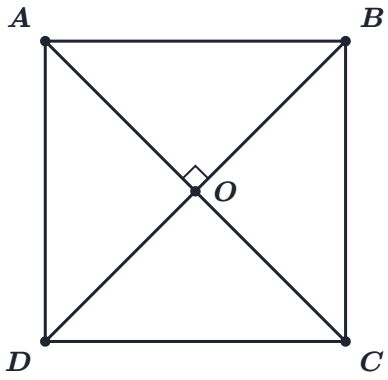
SOLUTION

The strips are the diagonals. They must be equal and bisect each other.

The other strip must also be 10 cm long. The angle between the strips can be anything.

3 Square

- A **square** is a quadrilateral with all angles 90° and all sides equal. Every square is a rectangle, but not every rectangle is a square.
- **Properties**: all the properties of a rectangle, and also the diagonals **bisect each other at** 90° and **bisect the angles** of the square (each half is 45°).
- To get a square from two strips: equal length, joined at their mid-points, **at right angles**.



Square: equal diagonals meeting at 90°

Example 3.1 The diagonal of a square is 8 cm. How would you construct the square without a protractor?

SOLUTION

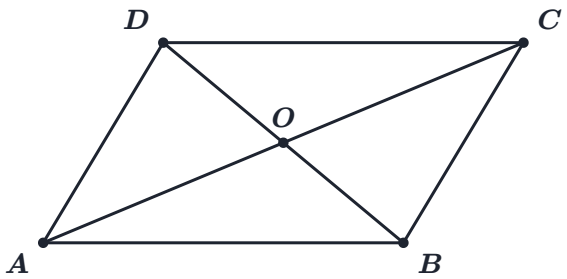
Draw $AC = 8$ cm and construct its perpendicular bisector, meeting AC at O.

Mark B and D on the bisector with $OB = OD = 4$ cm.

Join A, B, C, D. The diagonals are equal and bisect each other at right angles, so ABCD is a square.

4 Parallelogram

- A **parallelogram** is a quadrilateral in which both pairs of opposite sides are parallel.
- **Properties**: opposite sides are **equal**; opposite angles are **equal**; adjacent angles **add up to** $180°$; the diagonals **bisect each other**.
- Adjacent angles are interior angles on the same side of a transversal, which is why they add up to $180°$.
- A diagonal divides a parallelogram into two congruent triangles. That is how the equal sides are proved.
- The diagonals of a parallelogram need not be equal and need not be perpendicular.



Parallelogram: AO = OC and BO = OD

Example 4.1 In parallelogram ABCD, $\angle A = 60°$. Find the other angles.

SOLUTION

$\angle C = \angle A = 60°$ (opposite angles)

$\angle B = 180° - 60° = 120°$ (adjacent angles), and $\angle D = \angle B$

$\angle B = 120°, \angle C = 60°, \angle D = 120°$

Example 4.2 Two adjacent angles of a parallelogram are $(3x + 10)°$ and $(2x + 20)°$. Find all four angles.

SOLUTION

$3x + 10 + 2x + 20 = 180$, so $x = 30$

$100°, 80°, 100°, 80°$

Example 4.3 The diagonals of a parallelogram are 7 cm and 5 cm long. How far is each vertex from the point where they meet?

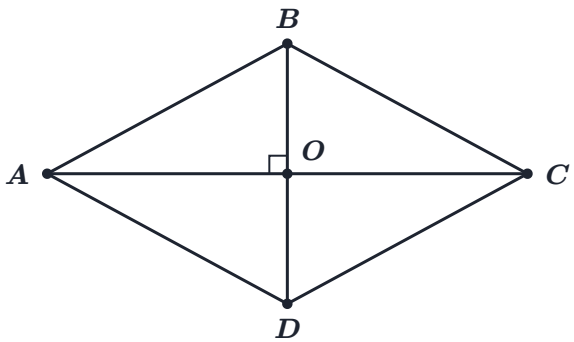
SOLUTION

The diagonals bisect each other.

3.5 cm, 2.5 cm, 3.5 cm and 2.5 cm

5 Rhombus

- A **rhombus** is a quadrilateral in which all four sides are equal.
- Every rhombus is a parallelogram, so it has all the properties of a parallelogram.
- **Extra properties**: the diagonals **bisect each other at** $90°$, and the diagonals **bisect the angles** of the rhombus.
- A rhombus with one right angle is a square.



Rhombus: the diagonals are perpendicular bisectors of each other

Example 5.1 One angle of a rhombus is $50°$. Find the other angles, and the angles a diagonal makes with the sides.

SOLUTION

Opposite angle $= 50°$. The other two are $180° - 50° = 130°$ each.

The diagonals bisect the angles.

Angles $50°, 130°, 50°, 130°$; the diagonals make angles of $25°$ and $65°$ with the sides

Example 5.2 The diagonals of a rhombus are 6 cm and 8 cm. Find its side.

SOLUTION

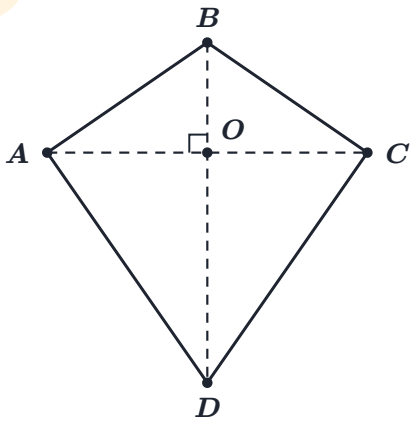
The half-diagonals 3 cm and 4 cm meet at right angles, so each side is the hypotenuse of a right triangle with legs 3 and 4.

Side$^2 = 3^2 + 4^2 = 25$

5 cm

6 Kite

- A **kite** is a quadrilateral ABCD with two pairs of equal **adjacent** sides: $AB = BC$ and $CD = DA$.
- **Properties**: the diagonal BD (joining the vertices between the equal sides) bisects $\angle B$ and $\angle D$, and it is the **perpendicular bisector** of the other diagonal AC.
- Reason: triangles ABD and CBD are congruent (SSS).
- Two congruent scalene triangles joined along a common side, as mirror images, form a kite.
- A kite that is also a parallelogram has all sides equal, so it is a rhombus.



Kite: AB = BC and CD = DA

Example 6.1 In kite ABCD, $AB = BC$, $CD = DA$, $\angle ABC = 100^\circ$ and $\angle ADC = 40^\circ$. Find $\angle A$ and $\angle C$.

SOLUTION

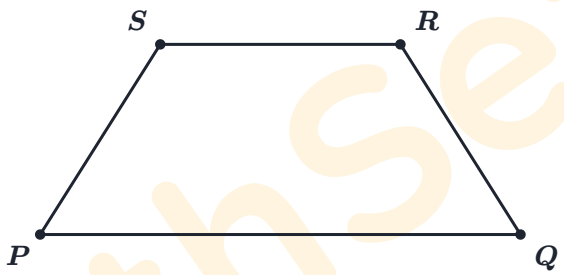
$\angle A = \angle C$ because triangles ABD and CBD are congruent.

$2\angle A = 360^\circ - 100^\circ - 40^\circ = 220^\circ$

$\angle A = \angle C = 110^\circ$

7 Trapezium

- A **trapezium** is a quadrilateral with at least one pair of parallel opposite sides.
- If $PQ \parallel SR$, then $\angle P + \angle S = 180^\circ$ and $\angle Q + \angle R = 180^\circ$.
- An **isosceles trapezium** has equal non-parallel sides. Its two base angles on the same parallel side are equal.



Isosceles trapezium: PQ is parallel to SR and PS = QR

Example 7.1 In trapezium PQRS, $PQ \parallel SR$, $\angle P = 75^\circ$ and $\angle Q = 45^\circ$. Find $\angle S$ and $\angle R$.

SOLUTION

$\angle S = 180^\circ - 75^\circ$ and $\angle R = 180^\circ - 45^\circ$

$\angle S = 105^\circ$ **and** $\angle R = 135^\circ$

Example 7.2 In an isosceles trapezium one base angle is 80°. Find all the angles.

SOLUTION

The other angle on the same base is also 80°. Each of the remaining two is $180^\circ - 80^\circ$.

$80^\circ, 80^\circ, 100^\circ, 100^\circ$

8 How the families are related

- Every square is a rectangle and a rhombus. Every rectangle and every rhombus is a parallelogram. Every rhombus is a kite. Every parallelogram is a trapezium.

	Parallelogram	Rectangle	Rhombus	Square
Opposite sides equal and parallel	Yes	Yes	Yes	Yes
All sides equal	No	No	Yes	Yes
All angles 90°	No	Yes	No	Yes
Diagonals bisect each other	Yes	Yes	Yes	Yes
Diagonals equal	No	Yes	No	Yes
Diagonals perpendicular	No	No	Yes	Yes
Diagonals bisect the angles	No	No	Yes	Yes

- **Recognising a quadrilateral from its diagonals**: bisect each other gives a parallelogram; equal and bisect each other gives a rectangle; bisect each other at right angles gives a rhombus; equal and bisect each other at right angles gives a square.
- A quadrilateral with both pairs of opposite sides equal, or with both pairs of opposite angles equal, is a parallelogram.

Example 8.1 The mid-points of the sides of a square are joined in order. What figure is formed?

SOLUTION

Each corner triangle is a right triangle with two equal legs (half the side), so the four new sides are equal and each corner triangle has angles $45^\circ, 45^\circ, 90^\circ$.

At each mid-point the angle of the new figure is $180^\circ - 45^\circ - 45^\circ = 90^\circ$.

A square

9 Common mistakes

- Saying a rhombus has equal diagonals, or that a rectangle has perpendicular diagonals. Only a square has both.
- Taking adjacent angles of a parallelogram as equal. They add up to 180°; the **opposite** angles are equal.
- Writing congruent triangles with vertices in the wrong order. The order must match the corresponding parts.
- Assuming perpendicular diagonals mean a rhombus. A kite also has perpendicular diagonals.
- Treating a measured result as proved. Measurement gives a conjecture; a proof needs reasoning.